

Problem Statement

What are we trying to solve?

Organizations are rapidly integrating AI tools for both leaders and employees, but there's limited understanding of how this affects their performance and well-being. While AI promises enhanced productivity, the relationship between AI adoption and workplace outcomes may be influenced by people's sense of competence. This study asks: **"How does AI integration affect employee and leader performance and well-being, and what role does perceived competence play in this relationship, considering the moderating effect of empowering leadership behavior?"**

Abstract

This study investigates the impact of AI integration on workplace performance and well-being among both leaders and employees, with a specific focus on how perceived competence mediates these relationships. Using validated psychological scales, we will examine whether AI adoption enhances or diminishes workplace outcomes, and how individuals' sense of competence influences these effects. Data will be collected from employees and leaders across industries using comprehensive surveys. The research aims to provide organizations with insights for effective AI implementation that enhances performance while maintaining employee well-being. Findings will help develop strategies for building competence during AI adoption.

Introduction

Why Study AI in the Workplace?

Artificial Intelligence (AI) is transforming work across organizational levels—from leadership decision-making to everyday employee tasks. While AI promises enhanced efficiency, understanding its psychological and performance impacts on both leaders and employees is crucial for successful implementation.

The Problem

Current research primarily focuses on technical aspects of AI implementation while overlooking:

- How AI adoption affects workplace performance across organizational levels
- The impact on employee and leader well-being
- The role of perceived competence in successful AI integration

Our Solution

This study examines two key outcomes of AI integration:

1. **Workplace Performance:** Using established scales to measure both leader and employee effectiveness
2. **Well-being:** Tracking psychological well-being in the context of AI adoption

We examine these relationships through the lens of:

- **Need for Competence:** How feeling capable and effective with AI influences outcomes

Why This Matters

Understanding these relationships will help organizations:

- Implement AI tools in ways that enhance rather than diminish performance
 - Support employee and leader well-being during AI adoption
 - Develop targeted interventions to build AI competence
 - Create more effective AI training programs
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Theoretical Framework

Key Hypotheses

H1: AI integration positively influences workplace performance

H2: AI integration positively influences well-being

H3: Need for competence mediates the relationship between AI integration and workplace outcomes

H4: The mediation effect of need for competence differs between leaders and employees

H5: Empowering leadership behavior moderates the relationship between AI integration and need for competence

H6: The moderating effect of empowering leadership behavior is stronger for employees than for leaders

Framework Diagram

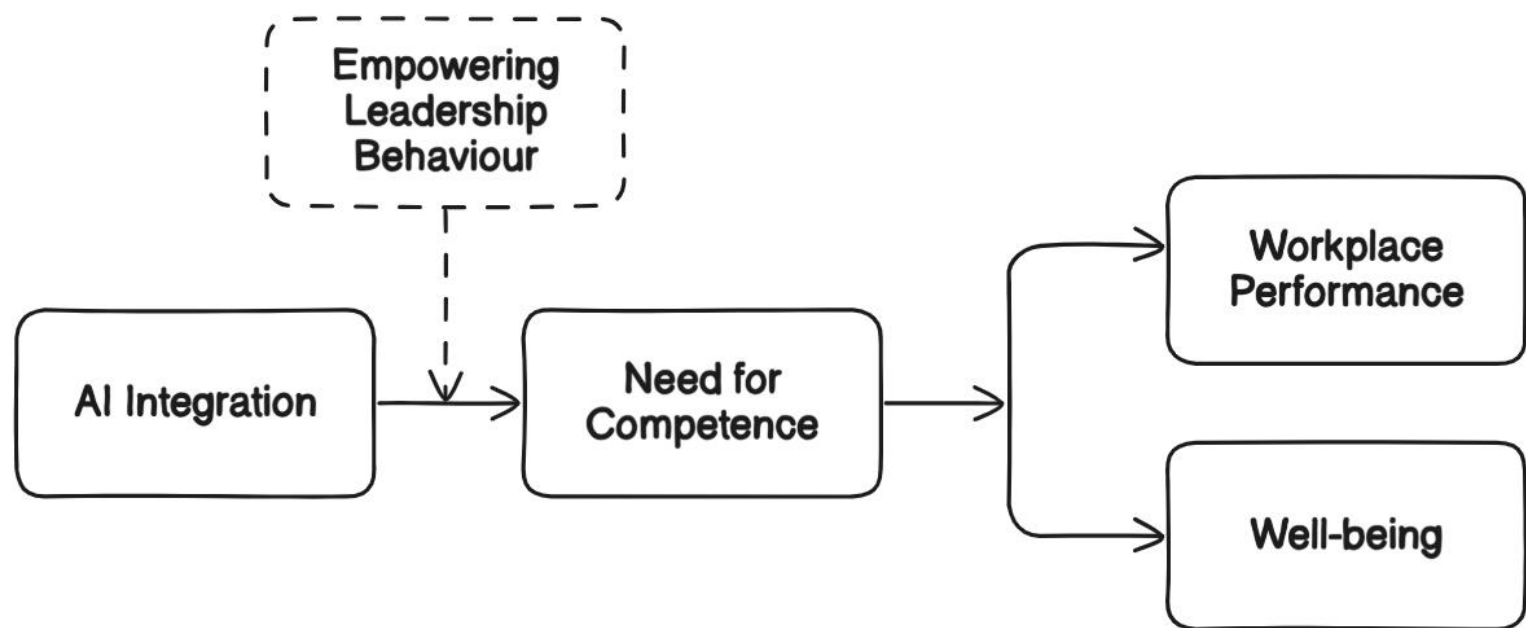


Figure 1: Framework Diagram

Methodology

1. Independent Variable: AI Integration

Scale: Technology Acceptance Model (TAM)

- Reference: Davis (1989)
- Alpha score: Alpha values ranging from 0.87 to 0.89 [1]
- Sample items:
 - "I find AI tools useful in my job"
 - "Using AI tools enables me to accomplish tasks more quickly"
 - "Learning to operate AI tools is easy for me"

2. Mediator: Need for Competence

Scale: Basic Psychological Need Satisfaction at Work Scale (BPNS-W)

- Reference: Deci et al. (2001)
- Alpha score: $\alpha > 0.85$ [2]
- Sample items:
 - "I feel capable in using AI tools effectively"
 - "I have been able to master new AI-related skills"
 - "I feel competent to achieve my goals with AI tools"

3. Moderator: Empowering Leadership Behavior

Scale: Empowering Leadership Questionnaire (ELQ)

- Reference: Arnold et al. (2000)
- Alpha score: $\alpha > 0.90$
- Sample items:
 - "My leader encourages me to solve problems independently"
 - "My leader provides me with the necessary resources for AI adoption"

- "My leader helps me understand how to use AI effectively"

4. Dependent Variables

Workplace Performance

Scale: Individual Work Performance Questionnaire (IWPQ)

- Reference: Koopmans et al. (2014)
- Alpha score: Alpha values ranging from 0.85 – 0.875 [3]
- Sample items:
 - "I was able to perform my work tasks effectively"
 - "I took initiative when AI tools presented new opportunities"
 - "I adapted well to changes in AI-related work tasks"

Well-being

Scale: WHO-5 Well-Being Index

- Reference: Bech et al. (1996)
- Alpha score: $\alpha = 0.87 - 0.90$ [4]
- Sample items:
 - "I have felt cheerful and in good spirits at work"
 - "I have felt calm and relaxed while using AI tools"
 - "My daily work life has been filled with things that interest me"

5. Control Variables

1. Role (Leader/Employee)
2. Years of AI experience
3. Age
4. Industry sector
5. Organization size

Survey Design

Table 1: Summary of scale items

Construts	Codes	Items
AI Integration	AI1	I find AI tools useful in my job
	AI2	Using AI tools enables me to accomplish tasks more quickly
	AI3	Using AI tools increases my productivity
	AI4	Learning to operate AI tools is easy for me
	AI5	My interaction with AI tools is clear and understandable
	AI6	I find AI tools easy to use
	AI7	I intend to use AI tools in my job
	AI8	I predict I will use AI tools in my job
Need for Competence	NC1	I feel capable in using AI tools effectively
	NC2	I have been able to master new AI-related skills
	NC3	I feel competent to achieve my goals with AI tools
	NC4	People at work tell me I am good at using AI tools
	NC5	I have been able to learn interesting new AI skills at work
	NC6	At work, I have opportunities to show how capable I am with AI
	NC7	I feel confident in my ability to learn new AI features
	NC8	I can handle the challenges that AI brings to my work
Empowering Leadership Behavior	ELB1	My leader encourages me to find solutions to AI-related challenges
	ELB2	My leader provides me with the necessary resources for AI adoption
	ELB3	My leader helps me develop my AI-related skills

	ELB4 ELB5 ELB6 ELB7 ELB8	My leader encourages me to learn from AI-related mistakes My leader supports my autonomous decision-making with AI tools My leader shares information about AI implementation plans My leader coaches me to improve my AI capabilities My leader recognizes my progress in using AI tools
Workplace Performance	WP1 WP2 WP3 WP4 WP5 WP6 WP7 WP8	I was able to perform my work tasks effectively using AI I took initiative when AI tools presented new opportunities I adapted well to changes in AI-related work tasks I collaborated effectively with others when using AI tools I kept up to date with AI-related changes in my work I came up with creative solutions using AI tools I took on extra responsibilities related to AI implementation My overall work quality improved with AI use
Well-being	WB1 WB2 WB3 WB4 WB5	I have felt cheerful and in good spirits at work I have felt calm and relaxed while using AI tools I have felt active and vigorous in my work I wake up feeling fresh and rested My daily work life has been filled with things that interest me
Control Variables	CV1 CV2 CV3 CV4 CV5	What is your current role How many years of experience do you have with AI tools? What is your age? What industry sector do you work in? What is the size of your organization?

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